

Transfer & Discharge Summary to Transplant Service

To: BMT Admissions — Dr. Elise Morgan

From: Hematology/Oncology Service

Date: 21.03.2024

Patient: **Brenda Campbell** (DOB 05.04.1954; MRN AML050)

Primary Diagnosis: Acute Myeloid Leukemia (AML) with **myelodysplasia-related mutations** — **RUNX1** and **SRSF2** mutated; ELN 2022 **Adverse** risk

Initial Treatment: **Venetoclax + Azacitidine** began **30.12.2022**

Reason for Transfer

Ms. Campbell has relapsed/refractory AML after an initial response to venetoclax/azacitidine with subsequent progression documented late August 2023. She has since received salvage therapy with intent to bridge to allogeneic hematopoietic cell transplantation. We are transferring her for transplant evaluation and, if deemed appropriate, admission for reduced-intensity conditioning.

Diagnostic Background

Presentation (23.12.2022): Several weeks of fatigue, bruising, and exertional dyspnea.

Admission labs: WBC **16.1 ×10⁹/L** (32% blasts), Hgb **7.6 g/dL**, Platelets **48 ×10⁹/L**; LDH **642 U/L**; uric acid **6.8 mg/dL**; fibrinogen 390 mg/dL; INR 1.1.

Bone marrow (23.12.2022): Hypercellular (80–90%) with **~60% myeloblasts**; multilineage dysplasia. Flow: CD34⁺, CD117⁺, HLA-DR⁺, CD13⁺/CD33⁺, aberrant CD7; MPO variable.

Cytogenetics: 46,XX; no core-binding factor or KMT2A rearrangement.

Molecular: **RUNX1** mutation (frameshift, VAF 36%), **SRSF2** p.P95 (VAF 31%), **ASXL1** subclonal (VAF 12%). **FLT3** and **NPM1** negative.

Baseline echocardiogram: LVEF **60%**. Hepatitis B/C, HIV serologies negative.

Comorbidities at baseline:

- Hypertension (lisinopril 10 mg qd).
- Stage **3a CKD** (baseline creatinine 1.2–1.4 mg/dL; eGFR ~45–55).
- Hypothyroidism (Hashimoto) — levothyroxine 75 µg qAM.
- Hyperlipidemia — atorvastatin 20 mg nightly.
- Osteoarthritis knees; uses acetaminophen; avoids NSAIDs.
- Former smoker (10 pack-years; quit 2008); drinks wine socially.

Allergies: NKDA.

Treatment Timeline & Responses

1) First-line: Venetoclax + Azacitidine

- **Start: 30.12.2022** — Azacitidine 75 mg/m² D1–7 + venetoclax with standard ramp to 400 mg daily D1–21 (cycles q28d).
- **Prophylaxis:** Acyclovir 400 mg bid; antibacterial during neutropenic nadirs; **azole antifungals avoided** (to preserve venetoclax dosing).
- **Complications:** Early cycles notable for grade 4 neutropenia and grade 3 thrombocytopenia; brief febrile neutropenia in Feb 2023 (culture negative). No tumor lysis.
- **Response assessments:**
 - **Marrow 27.02.2023:** Morphologic remission with **blasts 3–4%** (CRi), flow MRD low-positive (0.2%).
 - **Peripheral counts Spring 2023:** ANC ~0.9–1.2 ×10⁹/L; platelets 60–90 ×10⁹/L; transfusion needs decreasing.
 - **By July 2023:** Gradual count decline and rising LDH.
- **Relapse/Progression:**
 - **Bone marrow 28.08.2023: 28–30% blasts;** persistent RUNX1/SRSF2; ASXL1 VAF increased to 22%. Peripheral smear with circulating blasts (3–5%). Venetoclax-based therapy discontinued.

2) Salvage Strategy: Multidisciplinary decision to attempt **FLAG-IDA** (dose-attenuated for age/renal function) with **growth factor** support and aggressive anti-infective prophylaxis, aiming for rapid cytoreduction and transplant bridge.

- **Cycle 1 FLAG-IDA (05.09.2023):**
 - Fludarabine 25 mg/m² D1–5; Cytarabine 1.5 g/m² q24h D1–5 (renal-adjusted); Idarubicin 8 mg/m² D1–3; G-CSF priming.
 - **Toxicities:** FN requiring piperacillin-tazobactam→meropenem; mucositis grade 2. **AKI** peaked Cr 1.7 mg/dL, resolved with hydration and dose holds.
 - **Day ~28 marrow (03.10.2023): CRi** — blasts <5%, flow MRD **0.08%**.
- **Cycle 2 FLAG (no anthracycline) (02.11.2023):**
 - Given persistent MRD. Course complicated by **posaconazole-associated transaminitis** (ALT peak 112 U/L), switched to micafungin.
 - **Restaging 06.12.2023:** Flow MRD **0.04%**; RUNX1 mutation still detectable at low VAF (molecular MRD).

3) Pre-Transplant Consolidation & Bridge (Jan–Feb 2024):

- Two cycles **decitabine 20 mg/m² D1–5** with **low-dose cytarabine 20 mg bid D1–10**, chosen to maintain disease control with better tolerability while donor search completed.
 - **Donor: HLA 10/10 matched unrelated donor** identified (female, O+).
 - **Most recent marrow (04.03.2024):** Hypocellular, **blasts 2–3%**; flow MRD **0.03%**; **RUNX1 VAF 5%**, **SRSF2VAF 7%**. Platelets **88 ×10⁹/L**, ANC **1.0 ×10⁹/L**. Clinically well and motivated for transplant.
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Current Status (Pre-Transfer Clinic 18.03.2024)

Vitals/Exam: Afebrile; BP 122/68; HR 74; SpO₂ 99% RA; ECOG **1**. No lymphadenopathy; lungs clear; no HSM; scattered ecchymoses at venipuncture sites.

Laboratory data:

- **CBC:** WBC **3.6 ×10⁹/L** (ANC **1.1**), Hgb **10.6 g/dL**, Plt **94 ×10⁹/L**.
 - **CMP:** Na 138, K 4.3, **Cr 1.28 mg/dL** (eGFR 47), AST 34, ALT 41, Tbili 0.7, albumin 3.9.
 - **Coag:** INR 1.1; fibrinogen 360 mg/dL.
 - **Infectious screening:** CMV IgG+, EBV IgG+; PCRs negative.
- Imaging:** CT chest (02.2024) clear aside from mild basilar atelectasis.
Cardiopulmonary: Echo (02.2024) **LVEF 58%**; PFTs FEV₁ 78% predicted, DLCOcorr 72% (adequate for RIC).
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Medications on Transfer

- **Acyclovir 400 mg bid** (continue).
 - **Levofloxacin 500 mg qd** during neutropenia; stop if ANC >0.5 and per transplant protocol.
 - **Micafungin 100 mg IV daily** (bridge antifungal; prior transaminitis on azoles).
 - **Lisinopril 10 mg qd, levothyroxine 75 µg qAM, atorvastatin 20 mg qHS.**
 - **Acetaminophen** PRN; **ondansetron 8 mg q8h** PRN nausea; **senna** nightly PRN constipation.
 - Avoid NSAIDs given platelets <100.
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Central Line & Access

Right tunneled IJ catheter placed 09.2023; currently patent with no erythema. Last dressing change 17.03.2024. Recent blood cultures negative.

Transfusion History & Guidelines

She has required intermittent **RBC** (goal Hgb ≥ 8 g/dL) and **platelet** support (goal $\geq 10 \times 10^9/L$, ≥ 20 if bleeding). No alloimmunization detected; post-platelet increments adequate. Consider **CMV-safe, irradiated, leukoreduced** products exclusively.

Adverse Events / Special Considerations

- **Renal function:** CKD 3a; please avoid nephrotoxins and adjust conditioning/antimicrobials accordingly.
 - **Hepatic:** Prior azole transaminitis; voriconazole/posaconazole only with level monitoring or consider non-azole prophylaxis per your protocol.
 - **Infections:** No invasive fungal disease history; one FN episode each in Sept and Nov 2023 with negative cultures. Vaccinations up to date pre-transplant per inactivated schedule.
 - **Psychosocial:** Lives with spouse; strong support. Completed transplant education and advance directive; goal is curative-intent transplant.
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Assessment

AML with myelodysplasia-related gene mutations (**RUNX1**, **SRSF2**) initially responsive to venetoclax/azacitidine, with **biologically expected early relapse**. She achieved **CRi with low-level MRD** after dose-attenuated **FLAG-IDA**, maintained on low-intensity bridge while a **matched unrelated donor** was identified. Organ function and performance status are acceptable for **reduced-intensity allogeneic HSCT**.

Plan / Requests to Receiving Team

1. **Proceed with RIC allo-HSCT** using your institutional regimen (e.g., Flu/Bu or Flu/Mel) as soon as a bed is available. We recommend continuing **disease control** measures up to day -2 (your preference: maintenance decitabine D1-3 or observation if counts remain stable).
2. **GVHD prophylaxis:** Per your protocol (e.g., tacrolimus/methotrexate or PTCy-based).
3. **MRD tracking:** Please follow **RUNX1** (and **SRSF2**) VAF and flow MRD post-transplant at days +30, +100, +180 to guide pre-emptive therapy if transcripts rise.
4. **Infection prophylaxis:** Continue antiviral prophylaxis; consider mold-active prophylaxis post-engraftment if counts delayed; avoid azoles if liver enzymes climb.

5. **Cardio-renal protection:** Maintain $K^+ > 4.0$, $Mg^{2+} > 2.0$; renal-dose adjustments for calcineurin inhibitors and antimicrobials.

We will remain available for co-management and follow-up. Thank you for accepting Ms. Campbell for transplant evaluation.

Signed,

[Name], MD • Consultant Hematologist

Direct line: [redacted] | Nurse coordinator: [redacted]